Economics Letters 151 (2017) 91–95

Contents lists available at ScienceDirect

Economics Letters

journal homepage: www.elsevier.com/locate/ecolet

Revisiting risk aversion: Can risk preferences change with experience?

Eyal Ert [a], Ernan Haruvy [b,*]

[a] Department of Environmental Economics and Management, Robert H. Smith Faculty of Agriculture, Food and Environment, The Hebrew University of Jerusalem, P.O. Box 12, Rehovot, 76100, Israel

[b] Naveen Jindal School of Management, University of Texas at Dallas, Richardson, TX 75080, USA

HIGHLIGHTS

- We show that Holt–Laury preferences are not stable over repetitions.
- In a laboratory experiment, subjects repeatedly make choices with feedback in the Holt–Laury task.
- We test whether subjects adjust their responses with experience and if so in which direction.
- We find that subjects move towards payoff maximization with experience.

ARTICLE INFO

Article history:
Received 6 August 2016
Received in revised form
27 November 2016
Accepted 8 December 2016
Available online 19 December 2016

Keywords:
Risk aversion
Learning
Experiment

ABSTRACT

The Holt–Laury measure for risk aversion has been used extensively in economic studies to measure individuals' risk aversion. The idea behind this measure is that individuals have stable risk preferences when making decisions under risk. We show that having repeated experiences with the Holt–Laury task can move individuals from exhibiting "risk aversion" to displaying "risk neutrality." This finding suggests that either risk preferences are not robust to a few experiences or that responses to the tasks indicate something else. We show that a simple model of adaptation can capture this behavioral pattern.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

It is well documented that human decision makers are willing to sacrifice expected payoff in exchange for reduced risk. Risk aversion is a fundamental component of economic behavior (Friedman and Savage, 1948). There are alternative methods and approaches for measuring risk aversion, with nuances, advantages and disadvantages for each. For more comprehensive overviews, see Cox and Harrison (2008) and Charness et al. (2013). The most widely used measure of risk aversion is the Holt–Laury measure (Holt and Laury, 2002). This measure has been used widely to control for individual differences in risk preferences while assessing the effects of experimental manipulations (Brunner et al., 2014; McGee, 2013).

A fair question to ask is whether responses in the Holt–Laury task measure risk preferences. Anderson and Mellor (2009), for example, show that the Holt–Laury measure is uncorrelated with survey responses measuring risk taking behavior in various situations, indicating that it may not capture risk preferences that are manifested in risky behavior. A related question is whether responses in the Holt–Laury task are stable over time (Harrison et al., 2005). A body of work investigates *stability over time* by eliciting risk preferences at different dates (Andersen et al., 2008; Levin et al., 2007; Sahm, 2012). It is important to stress that the experimental results provided here, while temporal in nature, do not address the question of stability of risk preferences *over time*—but rather the question of stability of preferences *over repetitions*.

We show that Holt–Laury preferences are not stable over repetitions. We argue that this is a serious challenge to the interpretation of Holt–Laury responses as stable risk preferences. Specifically, we let subjects repeatedly make choices with feedback in the Holt–Laury task, with only a single decision drawn for payment. By doing so, we aim to test whether subjects adjust

* Corresponding author.
E-mail addresses: Eyal.Ert@mail.huji.ac.il (E. Ert), eharuvy@utdallas.edu (E. Haruvy).

http://dx.doi.org/10.1016/j.econlet.2016.12.008
0165-1765/© 2016 Elsevier B.V. All rights reserved.

Table 1
The ten paired lottery decisions.

Decision	Option A	Option B	$\Delta EV = B-A$
1	10/10 of \$2.00; 0/10 of \$1.60	10/10 of \$3.85; 0/10 of \$0.10	\$1.85
2	9/10 of \$2.00; 1/10 of \$1.60	9/10 of \$3.85; 1/10 of \$0.10	\$1.52
3	8/10 of \$2.00; 2/10 of \$1.60	8/10 of \$3.85; 2/10 of \$0.10	\$1.18
4	7/10 of \$2.00; 3/10 of \$1.60	7/10 of \$3.85; 3/10 of \$0.10	\$0.85
5	6/10 of \$2.00; 4/10 of \$1.60	6/10 of \$3.85; 4/10 of \$0.10	\$0.51
6	5/10 of \$2.00; 5/10 of \$1.60	5/10 of \$3.85; 5/10 of \$0.10	\$0.18
7	4/10 of \$2.00; 6/10 of \$1.60	4/10 of \$3.85; 6/10 of \$0.10	−\$0.16
8	3/10 of \$2.00; 7/10 of \$1.60	3/10 of \$3.85; 7/10 of \$0.10	−\$0.50
9	2/10 of \$2.00; 8/10 of \$1.60	2/10 of \$3.85; 8/10 of \$0.10	−\$0.83
10	1/10 of \$2.00; 9/10 of \$1.60	1/10 of \$3.85; 9/10 of \$0.10	−\$1.17

their responses with experience and if so in which direction.[1] With stable risk-preferences, experience should not have any effect on risk aversion because all information is known ex-ante. In other words, under the assumption that the Holt–Laury task captures stable risk preferences, subjects in theory have nothing to learn about the payoff distribution. In contrast, we find that subjects move towards payoff maximization, i.e., risk neutrality, with experience.

2. Experimental procedure

Sixty undergraduate students served as paid participants in the experiment. In each of 200 periods the subjects are presented with Holt and Laury's Multiple Price List (MPL) task (Table 1) and have to choose in each of the 10 decisions between option *A* and option *B*. After each period they receive feedback, which includes a realization of their choice and the outcome of the alternative option, about one randomly selected decision in that period. The last column in Table 1, not shown to subjects, displays the expected value differences between each pair of options. At the end of the experiment, they received payment according to the realization of their choice in one randomly chosen period.

3. Model predictions

We examine two competing theories in the current setting. One posits that subjects in the Holt–Laury task have stable preferences which can be captured by their ten decisions and indicate their risk preference. The other states that subjects are facing an unfamiliar environment and learn how to optimize in that environment over time with feedback.

3.1. The constant relative risk aversion model (CRRA) for a one-shot play

Holt and Laury posit that players engage in noisy expected value maximization with constant relative risk aversion (CRRA). Under this model any observed *A* choice before decision 7, where the riskier option *B* is associated with higher EV, is explained by risk aversion. Any observed *B* choice in decisions 7–10 can be attributed to risk seeking.

This explanation can be captured with two equations. Denote by $dec \in \{1, \ldots, 10\}$ the decision number in Holt–Laury's 10 decision task. Denote by $EU_{dec,j}$ the expected utility function for a two-outcome lottery j ($j = A, B$). Each lottery j has two possible outcomes; outcome 1 yielding payoff π_{j1}, and outcome 2 yielding payoff π_{j2}. Denote the probability for outcome 1 in lottery j by $p_{dec,j}$. Then the expected utility for lottery j is:

$$EU_{dec,j} = p_{dec,j}\pi_{j1}^{1-r} + (1 - p_{dec,j})\pi_{j2}^{1-r}. \quad (1)$$

This specification implies risk seeking for $r < 0$, risk neutrality for $r = 0$ and risk aversion for $r > 0$, and r is the *inherent risk preference* parameter to be estimated. The model assumes a probabilistic choice rule, which is typically implemented as a logit.

$$\Pr(\text{choose option } A \text{ in decision dec}) = \frac{\exp\left(\gamma EU_{dec,A}\right)}{\exp(\gamma EU_{dec,A}) + \exp(\gamma EU_{dec,B})}, \quad (2)$$

where γ is the "payoff sensitivity" parameter ($0 < \gamma < 1$) that determines the degree to which the differences in expected utilities between the two options' influence the probability of choice (Pr). Essentially, the model predicts that each of the lottery choices a player makes is independently made with *i.i.d.* errors. A decision maker is assumed to be more likely to choose an outcome which yields higher expected value than the alternative. When a player is observed to choose a lottery which yields in expectation lower expected payoff, it can be inferred that this is either due to error or there exists a mapping between payoffs and expected utility by which the chosen lottery is the higher expected utility choice, despite having a lower expected payoff than the alternatives.

Now let us take a close look at a commonly used variation on Eq. (1), used to model binary choice in general. Namely, in binary choice econometric modeling, it is common to assign an intercept to one of the choices, implying a non-equal propensity between the two choices when they are equal on all other dimensions.

$$EU_{dec,j} = \alpha(j = A) + p_{dec,j}\pi_{j1} + (1 - p_{dec,j})\pi_{j2}. \quad (1')$$

It turns out that the model of Eqs. (1), (2) and the model of Eqs. (1′), (2) are econometrically equivalent. That is, for every parametrization (r, γ) of (1) and (2), there exists a set of parameters (α, γ') for (1′) and (2) that yields the exact same choice probabilities and therefore the exact same likelihood.

We have not yet discussed any economic interpretation for α in model 2 that could be applied generally to economic modeling. One natural interpretation is that decision makers have an *inherent propensity* in the choice between safe and risky choices that is resistant to the expected payoffs computed from the instructions. We interpret α as the product of the weight on that inherent propensity and the difference between the alternatives implied by that propensity. If this inherent propensity story happens to describe people's behavior, then almost any feedback—even very limited feedback (as in the current experimental implementation) about payoffs would move people away from their otherwise seemingly risk-averse (or risk-seeking as is the case for some subjects) behavior towards payoff maximization.

[1] Recent evidence suggest that the provision of feedback might affect risk taking behavior even when people receive full description of the prospects they face (Erev et al., 2016).

3.2. Adaptive learning

There is a large body of literature on the best learning models for experimental data. The present work is not suited for shedding light on a fairly contentious learning model debate. Instead, we limit the scope of our discussion to the very restricted and basic argument that any proper[2] adaptive dynamic would, by definition, lead to payoff maximization over time. We use the standard fictitious play learning model (Brown, 1951) which serves as the most basic baseline in the literature and predates any ongoing contentious debate. The fictitious play model states that every observed outcome carries the same weight as any other outcome. Let $N_{dec,t-2}$ denote the number of observations of payoffs for decision *dec* up to and including period $t-2$. Let $D_{dec,t-1}$ be a dummy variable that takes on the value 1 if the payoff of decision *dec* is observed in period $t-1$ and 0 otherwise. Then the payoff expectation for period t is the weighted average of the expectation from period $t-1$, which was based on the previous $N_{dec,t-2}$ observations and the observed payoff in period $t-1$ if decision *dec*'s payoff was indeed observed in period $t-1$ (otherwise, the payoff expectation remains the same). Specifically:

$$E\pi_{dec,j,t} = \frac{E\pi_{dec,j,t-1}N_{dec,t-2} + D_{dec,t}\pi_{dec,j,t-1}}{N_{dec,t-2} + D_{dec,t-1}}. \quad (3)$$

The expected utility is then a slight modification of Eq. (1′) with the addition of a term for $E\pi_{dec,j,t}$ which enters linearly into expected utility:

$$EU_{dec,j} = \alpha(j = A) + p_{dec,j}\pi_{j1} + (1 - p_{dec,j})\pi_{j2} + E\pi_{dec,j,t}. \quad (4)$$

The interpretation of this equation is that expected utility is a weighted average of the inherent propensity, the true expected payoff (which subjects can compute from the payoff description) and an adaptive belief on payoffs based on repeated observations.

4. Data patterns and statistical tests

The main results, depicted in Fig. 1, reveal that the proportion of subjects choosing the safe choice – option *A* – in each of the decisions 1–6 (where the risky choice – option *B* – is better in expected value) decreases consistently from the first to the last block of 10 periods ($t(59) = 4.46, p < 0.001$).[3] However, in decisions 7–10, where the safer option *A* provides higher expected value, the proportion of subjects choosing option *A* slightly increases (albeit insignificantly, $t(59) = -0.72, p = 0.472$).

It is useful to also look at time trends. A linear regression with the number of safe choices as the dependent variable, and the block number as the explanatory variable yields a coefficient of negative 0.091 (Std. error 0.014, $p < 0.001$) for blocks of 10 periods. When a block is of size 10 periods, the coefficient is negative 0.019 (Std. error 0.003, $p < 0.001$). The two estimated coefficients are nearly exact rescaling of one another corresponding to the definition of the block size. This robustness to block definition suggests that a linear time trend a good descriptor of the data.

Taken together, the results show that on average subjects exhibit higher expected value maximization in the last block of periods than in the first block of periods, as well as "shift in risk preferences" from exhibiting "risk aversion" to exhibiting "risk neutrality". In other words, the results show that the Holt–Laury classification would classify the majority of subjects as risk averse in the first block of 10 periods, but the majority of subjects would be classified as risk-neutral in the last block of 10 periods by the same classification. This observation casts doubt on the stability of risk preferences as assessed by the Holt–Laury measure.

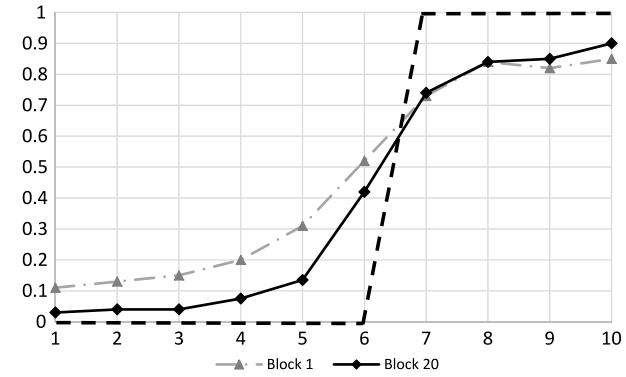


Fig. 1. Proportion of safe (A) choices in each decision in the first and last 10 periods. The dashed line represents the predicted choice of a risk-neutral player.

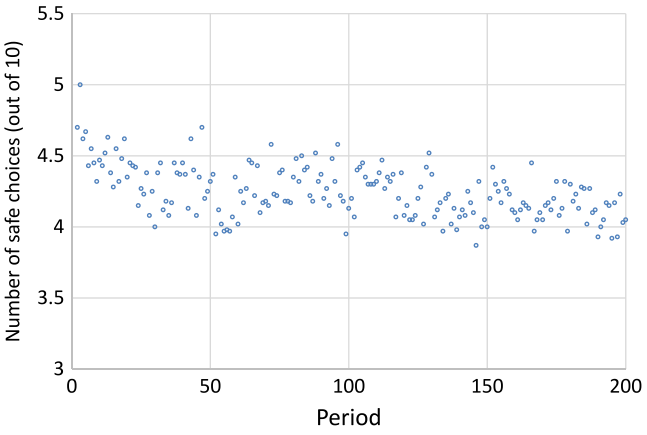


Fig. 2. Mean number of safe choices in each of the 200 periods.

Fig. 2 shows the learning process by depicting the mean number of safe choices in each of the 200 periods. The results indicate that in the early periods subjects choose, on average, 5 safe choices and thus are classified by Holt and Laury's "risk preference classification" as "slightly risk averse" (similar to their subjects in the low real payoff treatment). Yet Fig. 2 also show that subjects adjust their behavior and converge to choosing about four safe choices on average, which implies that they would be classified as "risk neutral" according to Holt and Laury's classification.

4.1. Model estimation of CRRA and inherent propensity

To evaluate parameter stability, we estimated both the CRRA model and inherent propensity model. Table 2 presents estimates for the two (equivalent) alternative models of one-shot play. Standard errors are shown in parentheses. All regressions are discrete choice logit regressions with the choice of the safe alternative as the dependent variable.[4] The number of observations

[2] A proper learning model (Fudenberg and Levine, 1998) is a model that converges asymptotically to payoff maximization.

[3] For this and other the next inference, we computed the average of decisions 1–6 per player for the first and last block and then ran a paired *t*-test to compare the two blocks.

[4] The probability of each observation dictated by Eq. (2). The maximized likelihood is the product of probabilities of the observed choices. Because we framed this in the standard economic representation of utilities, where the scaling parameter γ is multiplied by the constant α as well as by the *EU*, in order to get standard errors for the coefficients in our specification we have to resort to non-linear estimation (in SAS proc nlmixed). We verify statistical significance for clustered standard errors.

Table 2
Estimation of two alternative (equivalent) models of one-shot play. Standard errors in parentheses. All regressions are discrete choice logit regressions. The dependent variable is the choice of the safe alternative. $N = 600$ consisting of 10 one-period (first or last period) choices for 60 participants (clusters).

Coefficient	CRRA first period	CRRA last period	Inherent propensity first period	Inherent propensity last period
γ	2.103 (0.245)	2.310 (0.247)	1.477 (0.122)	1.887 (0.146)
r	0.239 (0.053)	0.139 (0.047)		
α			−0.290 (0.067)	−0.166 (0.057)
Log likelihood	−310.7	270.55	−310.7	270.55
BIC	634.2	553.9	634.2	553.9

Table 3
Estimation of a fictitious play model of repeated play. 60 participants (clusters) over 200 periods. Standard errors in parentheses. All regressions are discrete choice logit regressions. The dependent variable is the choice of the safe alternative. $N = 60 \times 200 \times 10$ consisting of 10 choices over 200 periods for 60 participants (clusters).

Coefficient	Estimate
α	−0.189 (0.008)
$E\pi_{dec,j,description}$	1.927 (0.016)
$E\pi_{dec,j,t}$	0.208 (0.011)
Log likelihood	−46,884
BIC	93,802

600 consisting of 10 one-period (first or last period) choices for 60 clusters (individual subjects).

The results, presented in Table 2, reveal that the estimates of the CRRA "payoff sensitivity" parameter γ are 2.103 and 2.310 in first and last period estimation of CRRA respectively. The differences are insignificant. In other words, the sensitivity to payoff differences is the same between first and last periods according to CRRA. However, and more importantly, the estimation of the CRRA risk preference parameter r drops from 0.239 (std. error 0.053) in first period estimation to 0.139 (0.047) in last period estimation, and this difference is statistically significant at $p < 0.01$. Therefore, if CRRA were the correct model of preferences, we would conclude that choosing repeatedly in Holt–Laury task shifted people's risk preferences from risk aversion far closer to risk neutrality.

Comparing the first and last period of the inherent propensity model gives a likewise inconsistent picture. By that model's estimation, subjects become more sensitive to payoff differences (γ increases) and their inherent propensity becomes far less important (α decreases in absolute magnitude). So we can reject both representations as measuring stable preferences over time.

4.2. Learning model estimation

An estimation of the *modified fictitious play model of repeated play*, discussed in Section 3.2, on the current data is shown in Table 3. In that model, expected utility is a weighted average of the inherent propensity, the true expected payoff (which subjects can compute from the payoff description) and an adaptive belief on payoffs based on repeated observations. The maximum likelihood estimation is done as described in Section 4.1.

We see that the estimate for the initial propensity α is in between the two estimates of the static model in Table 2, showing some consistency between the two sets of estimates. The coefficient on $E\pi_{dec,j,description}$, 1.927 (Std. error 0.016, $p < 0.01$) is likewise in the range of estimates for α identified in the static model in Table 2. The interest in the dynamic model is the coefficient on $E\pi_{dec,j,t}$. The estimates reveal that the effect of $E\pi_{dec,j,t}$ on $EU_{dec,j}$ is positive and significant at 0.208 (Std. error 0.011, $p < 0.01$) suggesting that subjects updated their behavior in accordance with the feedback they have received from their previous choices in the experiment, and this is the only explanation for the movement we see over repetitions.

5. Discussion

The Holt and Laury (2002) decision task is one of the most commonly used measures of risk preferences. Such application of the Holt–Laury task requires that the risk parameter measured by the task should be stable and robust to repeated choice and experience. We proposed a simple study that tests this prediction by letting subjects make repeated choices in the Holt–Laury task. We find that even though repeated experience does not provide new information about the underlying payoffs provided in the instructions, it seems to move subjects' behavior towards expected value maximization under risk neutrality. Therefore, although a static risk preference model cannot account for this behavior, a simple adaptive learning model is sufficient for capturing this pattern.

We ask why people change their behavior in the Holt–Laury task. The most direct explanation is that they simply adapt their behavior according to their experiences and payoff realizations (reinforcements) over time. A related learning explanation might be that people do not have rigid beliefs about probabilities to start out with, and so they adjust their probability beliefs over time by sampling outcomes repeatedly (Bradbury et al., 2015; Kaufmann et al., 2013). While the current research is not aimed to tease out competing learning explanations, future research may try disentangle different experiential mechanisms that may be at play. A potential critique of our approach might be that the conjecture that the repetitive choice design might have yielded boredom effects that in turn pushed people to take more risks over time. The current findings reject this conjecture, since the boredom assertion cannot explain why subjects become risk neutral over time and not risk seekers. Furthermore, this assertion also implies more noisy responses over time, and the data shows the opposite pattern (decreased variance) in accordance with learning mechanisms.

The finding that the Holt–Laury risk preference classification is not robust to experience poses a natural question regarding what it is that we really measure using Holt–Laury task in the lab. One potential answer is that rather than risk preferences we might be capturing noisy best response behavior that is subject to adaptation. To the extent that the Holt–Laury task does measure a preference, whatever that preference might be, the current study reveals that it is not stable, and could be modified even by the simplest form of experience.

Acknowledgment

This research was supported by a grant from the Israeli Science Foundation (ISF - 1739/14) to Eyal Ert.

References

Andersen, S., Harrison, G.W., Lau, M.I., Elisabet Rutström, E., 2008. Lost in state space: are preferences stable? Internat. Econom. Rev. 49, 1091–1112.
Anderson, L.R., Mellor, J.M., 2009. Are risk preferences stable? Comparing an experimental measure with a validated survey-based measure. J. Risk Uncertain. 39, 137–160.
Bradbury, M.A.S., Hens, T., Zeisberger, S., 2015. Improving investment decisions with simulated experience. Rev. Financ. 19, 1019–1052.

Brown, G.W., 1951. Iterative solution of games by fictitious play. Act. Anal. Prod. Alloc. 13, 374–376.
Brunner, C., Hu, A., Oechssler, J., 2014. Premium auctions and risk preferences: an experimental study. Games Econom. Behav. 87, 467–484.
Charness, G., Gneezy, U., Imas, A., 2013. Experimental methods: Eliciting risk preferences. J. Econ. Behav. Organ. 87, 43–51.
Cox, J.C., Harrison, G.W., 2008. Risk Aversion in Experiments. Emerald Group Publishing.
Erev, I., Ert, E., Plonsky, O., Cohen, D., Cohen, O., 2016. From Anomalies to Forecasts: A Choice Prediction Competition for Decisions under Risk and Ambiguity.
Friedman, M., Savage, L.J., 1948. The utility analysis of choices involving risk. J. Polit. Econ. 56, 279–304.
Fudenberg, D., Levine, D., 1998. Learning in games. Eur. Econ. Rev. 42, 631–639.
Harrison, G.W., Johnson, E., McInnes, M.M., Rutström, E.E., 2005. Temporal stability of estimates of risk aversion. Appl. Financ. Econ. Lett. 1, 31–35. http://dx.doi.org/10.1080/17446540420003115 76.
Holt, C.A., Laury, S.K., 2002. Risk aversion and incentive effects. Amer. Econ. Rev. 92, 1644–1655.
Kaufmann, C., Weber, M., Haisley, E., 2013. The role of experience sampling and graphical displays on one's investment risk appetite. Manage. Sci. 59, 323–340.
Levin, I.P., Hart, S.S., Weller, J.A., Harshman, L.A., 2007. Stability of choices in a risky decision-making task: a 3-year longitudinal study with children and adults. J. Behav. Decis. Mak. 20, 241–252.
McGee, P., 2013. Bidding in private-value auctions with uncertain values. Games Econom. Behav. 82, 312–326.
Sahm, C.R., 2012. How much does risk tolerance change? Quart. J. Finance 2, 1250020.